

Dynamical Systems

Many laws of nature can be stated as dynamical systems, more precisely using either differential equations or difference equations.

$$\text{I. } x: \mathbb{R} \rightarrow \mathbb{R}^n \quad t \mapsto x(t)$$



t is the time, a continuous time

$x(t)$ describes the state of a phenomenon that changes in time

A differential equation is a relation between $x(t)$ and its derivatives $x'(t)$, $x''(t)$, ... up to some order.

In mechanics $x(t)$ is the position of a material point, $x'(t)$ is its velocity, $x''(t)$ is the acceleration.

In chemistry $x(t)$ can be the quantity or the concentration of a chemical substance, $x'(t)$ is the rate of change.

In population dynamics $x(t)$ is the number or the density of individuals in a population (people, animals, fish ...) and $x'(t)$ is the rate of growth.

$$\text{II. } x: \mathbb{N} \rightarrow \mathbb{R}^n \quad k \mapsto x(k) = x_k \quad (x_k)_{k \geq 0} \text{ is a sequence}$$

k is the time, a discrete time

A difference equation is a relation between $x_k, x_{k+1}, x_{k+2}, \dots$

There are two main parts of dynamical systems theory

I Continuous dynamical systems

II Discrete dynamical systems

The aim of this theory is to make predictions for the evolution of a phenomenon when one knows the initial state and some "laws" of "nature". Mathematically this means that either we intend to find the expression of $x(t)$ or $(x_k)_{k \geq 0}$ or, if this is not possible, to be able to establish its properties.

properties.

A general form of a differential equation is $F(t, x(t), x'(t), \dots, x^{(n)}(t)) = 0$

1. Linear differential equations of order n , $n \in \mathbb{N}^*$

$$(1) \quad x^{(n)} + a_1(t)x^{(n-1)} + \dots + a_{n-1}(t)x' + a_n(t)x = f(t)$$

where $a_1, \dots, a_n : I \rightarrow \mathbb{R}$ continuous (called the coefficients) are given (known)
 $I \subset \mathbb{R}$ nonempty open interval

$f : I \rightarrow \mathbb{R}$ continuous (called the force or the nonhomogeneous part of (1)) is given (known)

The unknown is $x : I \rightarrow \mathbb{R}$ a scalar function.

Definition A solution of (1) is a function $\varphi : I \rightarrow \mathbb{R}$ which is n times differentiable and $\varphi^{(n)} : I \rightarrow \mathbb{R}$ is continuous such that

$$\varphi^{(n)}(t) + a_1(t)\varphi^{(n-1)}(t) + \dots + a_n(t)\varphi(t) = f(t), \quad \forall t \in I.$$

Notations Let $I \subset \mathbb{R}$ interval

$$C(I) = \{ f : I \rightarrow \mathbb{R} \text{ continuous} \}$$

$$C^n(I) = \{ f : I \rightarrow \mathbb{R} \text{ s.t. } \exists f, f', f'', \dots, f^{(n)} \text{ continuous on } I \}$$

So $a_1, \dots, a_n, f \in C(I)$ and a solution $\varphi \in C^n(I)$.

we consider the addition between two functions

$$\text{Let } f_1, f_2 \in C(I), \quad f_1 + f_2 \in C(I) \quad (f_1 + f_2)(t) = f_1(t) + f_2(t), \quad \forall t \in I$$

$$\alpha \in \mathbb{R} \text{ (scalar)}$$

$$\alpha \cdot f \in C(I)$$

$$(\alpha f)(t) = \alpha \cdot f(t) \quad \forall t \in I.$$

$C(I)$ is a vector space

$C^n(I)$ is a vector space.

when (1) is called linear?

why (1) is called linear?

For each $x \in C^n(I)$ we define the function $Lx : I \rightarrow \mathbb{R}$,
$$Lx(t) \stackrel{\text{def}}{=} x^{(n)}(t) + a_1(t)x^{(n-1)}(t) + \dots + a_n(t)x(t), \quad \forall t \in I.$$

Moreover, we define now $L : C^n(I) \rightarrow C(I)$
$$x \mapsto Lx$$

So, L is a map between two vector spaces.

Proposition L is a linear map, i.e.

$$L(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 L(x_1) + \alpha_2 L(x_2), \quad \forall x_1, x_2 \in C^n(I) \\ \forall \alpha_1, \alpha_2 \in \mathbb{R}.$$

Proof. Let $t \in I$. $L(\alpha_1 x_1 + \alpha_2 x_2)(t) =$

$$\begin{aligned} &= (\alpha_1 x_1 + \alpha_2 x_2)^{(n)}(t) + a_1(t)(\alpha_1 x_1 + \alpha_2 x_2)^{(n-1)}(t) + \dots + a_n(t)(\alpha_1 x_1 + \alpha_2 x_2)(t) = \\ &= \alpha_1 x_1^{(n)}(t) + \alpha_2 x_2^{(n)}(t) + \alpha_1(t) \alpha_1 x_1^{(n-1)}(t) + \alpha_1(t) \alpha_2 x_2^{(n-1)}(t) + \dots + \alpha_1(t) \alpha_1 x_1(t) + \alpha_1(t) \alpha_2 x_2(t) + \\ &= \alpha_1 Lx_1(t) + \alpha_2 Lx_2(t) \quad \blacksquare \end{aligned}$$

with this notation, eq (1) can be written as

$$(2) \quad L(x) = f \quad (\text{equation with } x \text{ as unknown})$$

A particular case of (2) is when $f = 0$, i.e.

$$(3) \quad L(x) = 0 \quad \text{this is called } \underline{\text{linear homogeneous dif. eq.}}$$

The set of solutions of (3) is called, from linear algebra course) the kernel of L , denoted $\text{Ker } L$.

When $f \neq 0$ eq (2) is called linear nonhomogeneous.

Examples. 1) $x + 2x' = 0$ first order differential eq.

$$x' + \frac{1}{2}x = 0$$

linear homogeneous, with constant coeff.

$x=0$ is a solution (in fact $x=0$ is a solution of any lin. hom. diff. eq.)

$$x=-2 \quad "(-2)' + \frac{1}{2}(-2) = 0" \Leftrightarrow "-1 = 0" \quad F$$

Thus $x=-2$ is not a solution.

$$x=a \quad (a \in \mathbb{R}) \quad \text{check} \quad " \frac{a}{2} = 0 " \quad \text{True} \Leftrightarrow a=0.$$

So, $x=0$ is the only constant solution of $x' + \frac{1}{2}x = 0$

$$\begin{aligned} \text{Try } x=at+b \quad x'=a \quad \text{check} \quad "a + \frac{1}{2}(at+b) = 0 \mid \forall t \in \mathbb{R} \Leftrightarrow \\ \Leftrightarrow "at + (2a+b) = 0 \mid \forall t \in \mathbb{R} \Leftrightarrow \begin{cases} a=0 \\ 2a+b=0 \end{cases} \Leftrightarrow a=b=0. \end{aligned}$$

$$x=0$$

$$\text{Try } x=e^{-\frac{t}{2}} \quad x' = -\frac{1}{2}e^{-\frac{t}{2}}$$

$$\text{check} \quad " -\frac{1}{2}e^{-\frac{t}{2}} + \frac{1}{2}e^{-\frac{t}{2}} = 0, \quad \forall t \in \mathbb{R} \quad \text{TRUE.}$$

So, $x=e^{-\frac{t}{2}}$ is another solution. In addition, $x=c e^{-\frac{t}{2}}$, $c \in \mathbb{R}$ arbitrary is also a solution.

2) $x'' + t x' = 3$ second order linear nonhomogeneous diff. eq.

coeff. is not constant
nonhom. part

3) $x \cdot x' = 0$ first order nonlinear differential eq.

Consequences of the linearity of $\mathcal{L}$ (The Linearity principle or the Superposition principle.)

$$(1) \quad \mathcal{L}(\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k) = \alpha_1 \mathcal{L}(x_1) + \alpha_2 \mathcal{L}(x_2) + \dots + \alpha_k \mathcal{L}(x_k) \\ \forall x_1, \dots, x_k \in C^n(I), \quad \forall \alpha_1, \dots, \alpha_k \in \mathbb{R} \quad \underline{k \geq 1, k \in \mathbb{N}}.$$

Proof by induction.

(2) If $x_1, \dots, x_k$ are solutions of the LHDE $\mathcal{L}(x) = 0$

then, $\forall \alpha_1, \dots, \alpha_k \in \mathbb{R}$ we have that $\alpha_1 x_1 + \dots + \alpha_k x_k$ is another solution of $\mathcal{L}(x) = 0$.

(3) Let $f_1, \dots, f_k \in C(I)$, let x_i be a sol. of $\mathcal{L}(x) = f_i$, $i = \overline{1, k}$

Let $\alpha_1, \dots, \alpha_k \in \mathbb{R}$. Then for the eq. $\mathcal{L}(x) = \alpha_1 f_1 + \dots + \alpha_k f_k$

we have the solution $\alpha_1 x_1 + \dots + \alpha_k x_k$.

For the proof of (2) and (3) we apply (1).

$x' + \frac{1}{2}x = 0$ | we know $x = c \cdot e^{-\frac{t}{2}}$ is a solution $\forall c \in \mathbb{R}$
 $\cdot e^{\frac{t}{2}}$ this is an integrating factor for this equation.

$$x' \cdot e^{\frac{t}{2}} + x \cdot \underbrace{\frac{1}{2} e^{\frac{t}{2}}}_{\left(e^{\frac{t}{2}}\right)'} = 0 \quad \forall t \in \mathbb{R} \quad (\Leftrightarrow) \quad \left(x \cdot e^{\frac{t}{2}}\right)' = 0 \quad (\Leftrightarrow)$$

$$x \cdot e^{\frac{t}{2}} = c \quad | \quad c \in \mathbb{R} \quad (\Leftrightarrow)$$

$$\boxed{x = c \cdot e^{-\frac{t}{2}}, \quad c \in \mathbb{R}}$$

Hence

There are no other solutions.

$x = c \cdot e^{-\frac{t}{2}}, \quad c \in \mathbb{R}$ is the general solution of our d.e.